Kebei Jiang

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Core Expertise

LLM Evaluation: LLM-as-a-judge calibration · Evaluation benchmarks · Synthetic data · Experiment design · Behavioral analysis
AI/ML: LLMs · VLMs · Model fine-tuning · RAG · Agentic AI
Technical: Python · PyTorch · Hugging Face · FastAPI · Docker · Azure · GCP

Experience

Microsoft *Jan 2020 – Sep 2026*
Senior Applied Scientist

- Developed and deployed LLM and VLM solutions spanning agentic AI, RAG, knowledge mining, and model fine-tuning.
- Designed evaluation methodologies and experiments for LLM-based agents and AI systems, including synthetic benchmarks, behavioral diagnostics, and quantitative performance analysis.
- Developed production ML/AI systems in Python, with experience in FastAPI, Docker, automated testing, and production monitoring.
- Led technical development across scientists, PMs, and customer engineers, including technical design, code reviews, implementation decisions, and customer engagements.

Viridien *Jan 2015 – Jan 2020*
Applied Scientist

- Applied statistical and machine-learning methods to large-scale seismic data.
- Developed CNN-based solutions for seismic interpretation.

Selected Research & Evaluation

Specification-to-Calibration: Synthetic Supervision for LLM Judges *2026*
Submitted to JUDGe @ NeurIPS 2026

- **Developed a method for automating LLM judge calibration:** compiled natural-language specifications into coverage-controlled positive/negative examples, using targeted behavioral contrasts to revise rubrics while holding the judge model fixed.
- Improved position-debiased accuracy by 3.58 ± 0.58 pp on LLMBar; feedback-conditioned and GEPA-based calibration also improved balanced accuracy on BigCodeBench.

Period-Closing Skill — Stateful, Multi-Turn Agent Evaluation *2026*

- Designed a benchmark of 30+ test cases covering diverse failure modes for an LLM agent that automates period-closing workflows involving due dates, calendars, worker assignment, and task dependencies; reviewed cases with the PM to validate representative scenarios.
- Built realistic multi-turn test environments with pre-existing state, calendar constraints, dependencies, and workers with similar names; fed simulated user responses back into the agent to test multi-turn behavior.
- Separated execution-trace quality from final task success and added trace/task verification across repeated runs.

Account Reconciliation Agent — Synthetic Benchmark & Evaluation *2026*

- Built an evaluation framework for an LLM agent resolving 440 reconciliation exceptions between ledger and subledger using synthetic accounting workflows with ground-truth resolutions and tool-use paths.
- Identified an unrealistic uniform distribution of matching cases, validated the expected distribution with the PM, and recalibrated synthetic data generation.
- Measured task success, tool-call accuracy/order/count, latency, and run-to-run consistency across repeated evaluations.

VLM Fine-Tuning — Controlled Diagnostic for Shortcut Learning *2025*

- Designed a hierarchical taxonomy spanning hundreds of fashion-item classes and aligned it with GPT-4o's understanding of retail products; generated ∼1M synthetic training examples and fine-tuned an open-source VLM with LoRA, SFT, and DPO.
- Designed a controlled diagnostic to investigate what the model learned: after updating the names of several classes in the taxonomy, the fine-tuned model failed to adopt the changes, exposing taxonomy memorization.
- Achieved ∼90% accuracy against GPT-4o ground truths; used the diagnostic to motivate separating domain adaptation from taxonomy alignment through distinct training strategies.

Multimodal RAG — Financial Insights Copilot *2024*

- Built an end-to-end RAG system for content-rich financial PDFs, spanning document parsing, multimodal indexing, retrieval, query decomposition, and knowledge-graph exploration.
- Identified unnecessary intermediate image storage as a document-processing bottleneck and redesigned the indexing flow around image bounding boxes; reduced parsing time by ∼80%.
- Systematically evaluated multiple BM25 configurations and reranking models, improved offline retrieval by ∼20%; the RAG work was released as part of Microsoft Financial Insights Copilot.

Education

Louisiana State University *2007 – 2014*
PhD, Physics

Publications

- Guang Yang, Kebei Jiang, David P. Lundquist, Prerna Singh. *Specification-to-Calibration: Synthetic Supervision for LLM Judges.* Submitted to JUDGe @ NeurIPS 2026.
- Moschoula Pternea, Prerna Singh, Abir Chakraborty, Yagna Oruganti, Mirco Milletari, Sayli Bapat, Kebei Jiang. *The RL/LLM Taxonomy Tree: Reviewing Synergies Between Reinforcement Learning and Large Language Models.* JAIR 2024. DOI: 10.1613/jair.1.15960.
- Yu Zeng, Kebei Jiang, Jie Chen. *Automatic Seismic Salt Interpretation with Deep Convolutional Neural Networks.* Proceedings 2019 3rd International Conference on Information System and Data Mining, pp. 16–20.

Patent

- Yinheng Li, Kebei Jiang. *Concurrent Labeling of Sequences of Words and Individual Words.* U.S. patent application, MS# 411939-US-NP, 2022.